Python Assignment
Topic-5: Looping (For, While)

Q1. Introduction to for and while loops.

Ans.

Loops are used to repeat a block of code multiple times. The 'for' loop is used when we know how many times we want to repeat the code. It works with sequences like list, string, or range. The 'while' loop is used when the number of repetitions is not known. It runs until the given condition becomes false. Loops help in reducing code repetition and make programs shorter. They are very useful for solving problems that require repeated actions.

Example:

```
for i in range(1,6):
   print(i)

x = 1
while x <= 5:
   print(x)
   x += 1
```

Q2. How loops work in Python.

Ans.

Loops work by running the same block of code again and again based on a condition. In a 'for' loop, Python takes each value from a sequence and executes the code. In a 'while' loop, Python checks the condition before each iteration. If the condition is true, the loop continues, otherwise it stops. Loops help in tasks like printing numbers, processing lists, and repeating operations. Proper use of loops makes code simple and efficient.

Example:

```
numbers = [2,4,6]
for n in numbers:
   print(n)
```

Q3. Using loops with collections (lists, tuples, etc.).

Ans.

Loops can be used with collections like lists, tuples, and sets to access each element one by one. This helps in performing operations on multiple values easily. It saves time and reduces the length of code. We can also use conditions inside loops to check values. This makes loops very powerful in handling data. It is a basic concept that every beginner must understand for writing programs.

Example:

```
Cars = ["BMW","AUDI","PORSCHE"]
for f in Cars:
   print(Cars)
```